When Verification Axes Can Support Scientific Claims: Identifiability, Attainability, and Non-Compensatory Promotion

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Abstract

Verification layers for learning-based research agents are often evaluated by benchmark scores, but a score can support a competence claim only when the target is identifiable, the compared interfaces can express the required outcome, and the panel is discriminating. We formalize these requirements through fibrewise Bayes error, a donor-factorization theorem, an outcome-adapter criterion, and data-processing and total-variation bounds. In a protected 420-case safety battery, a non-compensatory promotion relation made 0/360 false promotions versus 180/360 for the strongest frozen mechanism proxy, while both promoted 60/60 clean positives. The registered abstention contrast on that battery was saturated and remains a negative measurement result. On a distinct shortcut-resistant exact-axis battery, the governed pipeline selected the undetermined outcome on 30/30 eligible cases versus 0/30 for the same strongest proxy and 15/30 for the one escalation-capable mechanism; the paired difference against the strongest proxy was 1.0 (95% CI $[1.0, 1.0]$), while the margin against the 15/30 mechanism was 0.5. This measures outcome expressiveness under the frozen gate lattice, not general scientific judgement. On 400 exact contracts, the target-bound relation scored 400/400 versus 250/400 for a donor-complete generic product and 50/400 for a compensatory product, while an information-equivalent typed product tied at 400/400. The paired improvement over the donor-complete generic product was 0.375 with a domain-stratified 95% bootstrap interval of $[0.3275, 0.4225]$. These finite results support an identifiability-and-attainability boundary for scientific promotion; they do not establish naturalistic or deployed-system superiority.

Keywords: scientific verification; benchmark identifiability; attainable error; provenance; authorization; protected evaluation; abstention.

1 Introduction

Language-model research systems increasingly retrieve sources, synthesize claims, attach citations, and run automated verification (Rasheed et al., 2026; Alvarez et al., 2026). This creates a governance problem: a claim may be correct but attributed to the wrong source; a source may support a nearby claim but not the assigned one; a checker may be non-empty yet non-discriminating or share lineage with the answer; and an evaluator may be contaminated or tampered with after the fact. In each case, a scalar confidence score can remain high even though a prerequisite for scientific authority is absent.

We study a stricter object: a *scientific-authority transition*. A proposal can be plausible and well supported but is promoted only when a registered set of content, provenance, checker, and evaluator prerequisites all pass. Unresolved prerequisites do not contribute a low score; they prevent promotion and leave the decision undetermined. This design is motivated by source-aware verification, multi-source attribution, claim-level auditability, evidence influence, iterative fact checking, evaluator-tampering benchmarks, and search-time

A large language model was used to assist with manuscript editing and preparation of the review package.

contamination work (Alvarez et al., 2026; Li et al., 2024; Rasheed et al., 2026; Faizan & Alharthi, 2026; Xie et al., 2024; Atinafu & Cohen, 2026; Wang et al., 2026c; Sadeghi et al., 2026). Recent academic-integrity and agentic-abstention benchmarks also make explicit that declining to complete an unsupported action can itself be the correct behavior rather than a failure to optimize task success (Yang et al., 2026; Liu et al., 2026).

The contribution is deliberately narrower than those component techniques. We ask whether composing these requirements as a *non-compensatory, protected authority transition* materially reduces false scientific-authority promotion while preserving legitimate clean coverage. Throughout, the pipeline built to that rule is called the *governed pipeline*, so that each claim reads as a claim about the rule rather than about a product; Section 5 names the implementation that instantiates it.

The publication-authorizing campaign was executed only after the subject, comparator and ablation mechanisms, evaluator, statistical rules, and split-generation code were frozen. A separate 39-case live-model arm remains exploratory because post-run audit found label/denominator inconsistencies and a hidden-family execution leak; it is retained as adverse evidence and is not used to authorize, tune, or rescue the protected result.

Our main findings are: (i) the governed pipeline made 0/360 false promotions while the strongest frozen mechanism proxy made 180/360, with the paired interval entirely beyond the predeclared five-point margin; (ii) both promoted 60/60 clean positives, satisfying the clean-coverage non-inferiority safeguard; (iii) the registered undetermined-outcome contrast was not supported on that battery because every system was correct on a structurally saturated family, whereas a shortcut-resistant battery supported the same contrast at 1.0 (95% CI $[1.0, 1.0]$) against the strongest comparator's 0/30 and at 0.5 against the 15/30 of the one comparator mechanism that can escalate; this contrast measures the ability to express an undetermined outcome rather than finer-grained judgement; and (iv) every registered ablation increased false promotion without reducing clean coverage, with soft confidence producing the largest degradation.

2 Related work

Source ownership and attribution. Attribution evaluation across multiple candidate sources is an established measurement problem (Li et al., 2024); source-aware factuality work targets cross-source conflation directly (Alvarez et al., 2026); and scientific claim–evidence benchmarks test whether retrieved evidence supports or contradicts a claim (Javaji et al., 2025). Claim-source retrieval sharpens the same point from the retrieval side, since recovering the source a scientific statement actually came from is a different task from finding a source that would support it (Schreieder et al., 2026). Together these establish claim correctness, source ownership, and semantic support as separate coordinates, which is the premise this paper builds on.

Provenance-sensitive authorization. The distinction between relevant evidence and evidence authorized to determine an action or conclusion is already established. A controlled audit holds the task and proposition fixed while varying source authority and measures whether agent actions change (Liao, 2026). Partial Evidence Bench studies a complementary failure mode in which access control is correct but a system overstates completeness when material evidence lies outside its authorization boundary (Tallam, 2026). AuthGraph aligns execution provenance against a separately derived authorization graph at tool and parameter-source level (Wang et al., 2026b); FAVA lowers structured task constraints into evidence-backed permission graphs checked before effectful execution (Zhang et al., 2026); and cryptographically verifiable authorization formalizes bindings among principal, request, context, policy, and execution evidence (Llambí-Morillas & Fernández-Fernández, 2026). These works absorb any generic claim that provenance-aware authorization, permission graphs, or evidence-backed authorization are new here. The residual question is narrower: when the target is *scientific claim promotion*, what representation and output interface make that target attainable, and does a target-bound non-compensatory relation differ from a donor-complete generic authorization relation on prospectively frozen exact contracts?

Auditability, citation fidelity, and influence. Claim-level auditability work argues that research-agent outputs require persistent semantic provenance rather than only fluent text (Rasheed et al., 2026). ProvenAI separates citation fidelity from whether cited evidence influenced answer generation (Faizan & Alharthi, 2026). We adopt that separation: citation presence cannot substitute for evidence-use provenance.

Research-integrity provenance and declared behavior. INSPECT-AI and RIPE-KG represent the provenance of research-integrity assessments over clinical-trial publications, showing a current scientific setting in which the assessment process itself is an auditable object (Markovic et al., 2026). Behavioral Integrity Verification formalizes a complementary declared-versus-actual integrity problem for agent skills (Wu et al., 2026), and certified speculative execution shows the same instinct applied to untrusted agent actions: let the work proceed, but hold its effects behind a check that must clear before they become authoritative (Zhou et al., 2026). These are parent-domain precedents for explicit provenance and contract checking.

Iterative verification and evidence escalation. FIRE iterates retrieval and verification rather than treating a single retrieval pass as final (Xie et al., 2024). DeepSciVerify motivates escalation from an abstract-level judgment to fuller evidence when initial evidence is insufficient (Sadeghi et al., 2026). Stage-wise fact-checking benchmarks make the complementary measurement argument, that an end-to-end verdict hides which stage actually failed (Lin et al., 2026). The first two mechanisms inform frozen comparator proxies here, and the third informs the decision to report per-family and per-coordinate outcomes rather than one aggregate.

Evaluator integrity, benchmark auditing, and contamination. RewardHackingAgents treats evaluator manipulation and held-out leakage as an explicit benchmark dimension (Atinafu & Cohen, 2026). Search-time contamination work shows that browsing systems can retrieve public benchmark answers during inference (Wang et al., 2026c). BenchGuard and Automated Benchmark Auditing show that benchmark specifications, environments, ground truths, and grading logic can themselves be defective (Tu et al., 2026; Wang et al., 2026a). The Holistic Agent Leaderboard further demonstrates the value of standardized harnessing and trace inspection, including direct observation of agents searching for benchmark material rather than solving the intended task (Kapoor et al., 2025). These results motivate the protected evaluator identity, holdout custody, access telemetry, and independent audit used here.

Abstention and scientific integrity. SciIntegrity-Bench constructs academic-integrity dilemmas in which honest acknowledgement of inability is the only correct outcome, while AgentAbstain evaluates paired should-act/should-abstain tasks in executable environments (Yang et al., 2026; Liu et al., 2026). They establish abstention as an evaluation target distinct from generic task success. The object measured here is narrower: an undetermined or blocked terminal is authority-specific, produced by an unresolved or failed evidence, checker, or evaluator prerequisite rather than by a general judgement that the task should not be attempted.

3 Verification-axis identifiability and attainability

A benchmark does not measure a scientific competence merely because it assigns a label and reports an accuracy. The label must be recoverable from the intended scientific information, unavailable from construction nuisances, expressible by the systems being compared, and capable of varying over the comparison panel. These are different requirements. We formalize them before interpreting the protected results.

Let $Y_a \in \mathcal{T}$ be the correct outcome on verification axis a for a random benchmark unit, let D be the representation available to a decision rule, and let finite nonempty $\mathcal{A} \subseteq \mathcal{T}$ be the outcomes that the rule is allowed to emit. For a finite benchmark distribution, the smallest attainable exact-outcome error is

$$\mathcal{E}_a^{\min}(D, \mathcal{A}) = 1 - \mathbb{E}_D \left[\max_{t \in \mathcal{A}} \Pr(Y_a = t \mid D) \right]. \tag{1}$$

This quantity separates a poor implementation from a task that its information or interface makes impossible.

Proposition 1 (exact axis attainability). There exists a decision rule $g(D) \in \mathcal{A}$ with zero exact-outcome error if and only if every positive-probability fibre of D contains one target outcome and that outcome belongs to $\mathcal{A}$. Equivalently, $Y_a = g(D)$ almost surely for some g with codomain $\mathcal{A}$. This is an almost-sure benchmark statement: it says nothing pointwise about worlds on null fibres.

Proof. On a fibre $D = d$, any rule must emit one outcome. It is correct on the whole fibre exactly when the conditional support of Y_a is the singleton containing that outcome. Summing the best fibrewise choices gives Equation 1, and zero risk gives the stated condition. □

Corollary 1 (donor-product factorization at two authorities). Let D collect the states exposed by a product of provenance, verification, custody, freshness, and generic-authorization donors, and let Y_{promote} be the target scientific-promotion outcome. Pointwise extensional equivalence on a declared world class holds if and only if the target outcome is constant on every donor-state fibre in that class and belongs to the output alphabet. Zero error under a benchmark distribution requires only the corresponding condition on positive-probability fibres; null fibres cannot be certified by a distributional score. If a positive-probability fibre mixes target outcomes, every donor-only rule has error at least

$$\sum_d \Pr(D = d)\left[1 - \max_{t\in\mathcal{A}} \Pr(Y_{\text{promote}} = t \mid D = d)\right]. \tag{2}$$

The pointwise statement is the set-theoretic factorization argument; the distributional statement is Proposition 1, and the displayed lower bound is attained by a fibrewise Bayes rule on the finite benchmark. Thus a separate module has no inherent advantage once a donor product receives a target-sufficient representation and the same decision relation; conversely, no optimization of a representation that merges differently licensed states can recover the missing distinction. This is a factorization boundary, not a claim that scientific authority must be centralized.

Corollary 2 (terminal-preserving comparator adapters). Let $V \in \mathcal{V}$ be the complete candidate-visible record supplied to an external comparator and let $X \in \mathcal{X}$ be its native output, where $\mathcal{V}$ and $\mathcal{X}$ are standard-Borel spaces. Allow X to be generated by a measurable randomized kernel from V, so that the comparator output adds no world information beyond V. A benchmark adapter is a prospectively declared measurable map $h : \mathcal{X} \to \mathcal{A}$; it is not a post-hoc relabelling rule. Its exact-terminal risk is bounded below by

$$\mathcal{E}_a^{\min}(X, \mathcal{A}) = 1 - \mathbb{E}_X\left[\max_{t\in\mathcal{A}} \Pr(Y_a = t \mid X)\right] \;\geq\; \mathcal{E}_a^{\min}(V, \mathcal{A}).$$

The first bound is attained by a measurable fixed-order Bayes adapter, and the second inequality is the data-processing consequence of $Y_a - V - X$. Thus a native output is admissible for a three-way exact-axis endpoint with zero benchmark error if and only if its positive-probability output fibres are target-pure and their targets belong to $\mathcal{A}$. For a deterministic native-output map, pointwise admissibility on a declared world class requires the same purity on every native-output fibre, including benchmark-null fibres. In particular, a binary native alphabet cannot attain an endpoint on which all three target outcomes have positive probability, and a native blocked, "not attributable," parse failure, or free-text insufficiency response cannot be recoded as undetermined unless the predeclared semantics-preserving map satisfies this fibre condition. When it does not, the mixed-fibre risk is an interface-attainability result and must not be reported as comparator inferiority.

Proposition 2 (claim-level indistinguishability). Let W_0 and W_1 be two scientific worlds in which a proposed competence claim has opposite truth values, and let O be the complete observable record available to an adjudicator. Write $P_i = \mathcal{L}(O \mid W_i)$ for probability measures on one common measurable record space $(\mathcal{O}, \mathcal{F}_O)$, and use the convention $\mathrm{TV}(P_0, P_1) = \sup_{B\in\mathcal{F}_O} |P_0(B) - P_1(B)|$. Under equal prior weight, every measurable, possibly randomized adjudicator has error at least

$$\frac{1 - \mathrm{TV}(\mathcal{L}(O \mid W_0), \mathcal{L}(O \mid W_1))}{2}. \tag{3}$$

The infimum error over adjudicators equals this bound. For two probability measures it is attained: a Hahn decomposition of the finite signed measure $P_1 - P_0$ supplies a maximizing acceptance region. In particular, if the observable records have the same distribution, the competence claim is not identifiable and the error is exactly 1/2 for every adjudicator.

Proof. For any acceptance region B, the equal-prior error is $\{P_0(B)+P_1(B^c)\}/2$. Taking the infimum over B gives one half of one minus the total-variation distance. The positive set in a Hahn decomposition of $P_1 - P_0$ attains the supremum, while randomized acceptance probabilities cannot improve on it because their risks are mixtures of deterministic risks. □

This second bound concerns interpretation rather than outcome prediction. A score can separate output interfaces while leaving two explanations of that score observationally equivalent. For example, one rule may recognize an epistemic insufficiency, whereas another may be forced into the only available non-promoting outcome. A broader competence headline remains unidentified until a discriminator changes the observable law of those worlds.

Nuisance identifiability. Let N contain construction-only quantities such as list length, missingness, template, key order, record-layout marker, or label-bearing metadata. The nuisance-only Bayes advantage is

$$I_a(N) = \mathbb{E}_N\left[\max_t \Pr(Y_a = t \mid N)\right] - \max_t \Pr(Y_a = t). \tag{4}$$

A scientific interpretation requires the intended semantic representation to make the outcome attainable while nuisance information contributes no material advantage. A finite shortcut register cannot certify $I_a(N) = 0$ for every possible decoder. It certifies only that the frozen probe class found no such advantage, which is why claim-axis authorization must name the probe class and cannot be promoted to whole-register or naturalistic clearance.

Panel resolution and interface attainability. For a frozen panel $\mathcal{H}$ and score M_a, define its observed resolution as

$$\rho_a(\mathcal{H}) = \max_{h\in\mathcal{H}} M_a(h) - \min_{h\in\mathcal{H}} M_a(h).$$

When $\rho_a = 0$, the experiment supplies no comparative ordering on that axis; it does not establish equality of the underlying competences. When $\rho_a > 0$ but some comparators lack a target outcome in their output alphabet, the contrast establishes interface expressiveness before it establishes the finer judgement of when that terminal is scientifically appropriate. A broad verification-axis claim is therefore licensed only inside the intersection of four conditions: semantic attainability, nuisance resistance, matched terminal attainability, and nonzero panel resolution. Failures fall into distinct research programmes rather than one generic null: repair the construction, enlarge the interface, expose a target-sufficient representation, or improve the decision rule.

Application to the three empirical layers. The primary safety battery's undetermined-outcome axis had $\rho_a = 0$: an empty evidence list exposed the gold family and every panel member scored 30/30. That comparison remains not supported. The later shortcut-resistant construction removes the registered cues and obtains nonzero resolution, but nine of ten comparator mechanisms cannot emit an undetermined outcome; its exact claim is therefore outcome expressiveness under the frozen gate lattice, not general epistemic judgement.

Stated as an estimand: where a comparator interface cannot emit an undetermined outcome, the quantity this comparison estimates is *outcome attainability* on the axis, and nothing else. A system that cannot express the correct outcome cannot be observed to reach it, so a gap measured against such an interface is a fact about what the interface can say, not about what the system knows. Reading it as epistemic competence would attribute to judgement what belongs to expressiveness. The exact-contract extension tests the donor-factorization boundary more directly. Its information-equivalent typed donor product receives the target-sufficient coordinates and predicate and ties exactly at 400/400, while the donor-complete generic and compensatory products do not. These layers diagnose different limits and are not pooled.

Table 1: Protected safety battery and custody. Hidden labels remained evaluator-only until all mechanisms were scored.

Slice	Cases	Clean	Hostile	Candidate labels
Post-run public slice	150	30	120	hidden during execution
Protected hostile	120	0	120	hidden
Protected holdout	150	30	120	hidden
Total	420	60	360	hidden

4 Non-escalating authority transition

Let a proposal contain claim c and evidence references $E = \{e_1, \ldots, e_n\}$. Promotion is a conjunction of hard prerequisites rather than a weighted score:

$$G_{\text{content}} : \text{resolved evidence exactly matches the registered content,} \tag{5}$$
$$G_{\text{owner}} : \text{the assigned source owns the cited content,} \tag{6}$$
$$G_{\text{support}} : \text{the assigned evidence supports } c, \tag{7}$$
$$G_{\text{indep}} : \text{checker lineage is admissibly independent,} \tag{8}$$
$$G_{\text{disc}} : \text{checker rejects the frozen hostile battery,} \tag{9}$$
$$G_{\text{influence}} : \text{cited evidence participated in answer generation,} \tag{10}$$
$$G_{\text{eval}} : \text{evaluator identity and evaluation order are intact,} \tag{11}$$
$$G_{\text{access}} : \text{protected-label access telemetry is clear,} \tag{12}$$
$$G_{\text{search}} : \text{search-time contamination is clear.} \tag{13}$$

The claim is promoted only when every required gate passes and the evidence supports it. A resolved negative prerequisite blocks promotion; an unresolved evidential prerequisite leaves the claim undetermined. No failed coordinate can be averaged away by confidence elsewhere. Eight ablations remove one coordinate at a time, including a soft-confidence variant that promotes when at least six of nine gates pass.

5 Methods

5.1 Three-valued outcomes

Every authority decision was three-valued rather than binary: a claim could be promoted, blocked, or left *undetermined*. An undetermined outcome was recorded separately from an outcome determined to be false, and neither category was merged with the other. This distinction makes a missing prerequisite observable without treating the claim as refuted.

5.2 Prospective design and protected split

The study protocol, comparator mechanisms, ablations, evaluator, statistical rules, and split-generation procedure were fixed before outcome-bearing evaluation. A public-seed preflight exercised 42 hostile authority tests and the complete 420-case procedure with ten comparator mechanisms, eight ablations, five deterministic repeats, protected scoring, typed-panel validation, and a separately implemented checker. A subsequent isolated evaluation generated a fresh secret-seeded split; the seed was unavailable to every scored mechanism.

The final split contains 420 mechanical-gold cases across 13 families: 60 clean positives and 30 cases in each of 12 hostile or insufficient-evidence families. A 150-case post-evaluation public slice contains 30 clean and 120 hostile cases, while distinct hostile and holdout slices remained protected during scoring. Table 1 summarizes the design.

5.3 Gold-blind evaluation and custody

Each scored mechanism received only an opaque case identifier and the candidate-visible evidence packet. Before evaluation, an automated check confirmed that attack family, protected gold, custody class, and expected outcome were absent. Candidate and comparator mechanisms ran in isolated environments without access to study source materials. Five deterministic repeats were monitored for protected-identifier disclosure and external network connections. The evaluator accessed gold only after all scored mechanisms had finished, and a separately implemented checker recomputed the headline counts. These controls establish gold isolation and implementation separation within one project; they do not constitute institutionally independent evaluation or external replication.

5.4 Comparator panel, ablations, and statistics

Seven primary comparator mechanisms mirror published ideas: ProvenanceGuard-style source routing, AttributionBench-style multi-source attribution, FIRE-style iterative retrieve/verify, CLAIM-BENCH-style claim evidence, ProvenAI-style citation fidelity/influence, RewardHackingAgents-style contamination/tamper checks, and DeepSciVerify-style evidence escalation. Three auxiliary controls test citation presence, pooled semantic support, and auditability/provenance.

These are protocol-matched reimplementations under one candidate-visible packet and common accounting, not executions of the cited authors' external software.

The eight registered ablations remove, one at a time, exact content binding, source/provenance distinction, checker-lineage independence, hostile-checker discrimination, behavioral influence, evaluator protection/telemetry, or the search-contamination block, or replace the fail-closed rule with a six-of-nine soft-confidence threshold.

The primary endpoint is false authority-promotion rate over 360 gold outcomes that do not permit promotion. The primary hypothesis requires an absolute improvement beyond 0.05 against the strongest frozen comparator. Clean coverage is the promotion rate over 60 clean positives and is required to be non-inferior within 0.05. A third registered contrast tests correct selection of the undetermined outcome. Rate intervals use Wilson 95% intervals; paired effects use a fixed-seed percentile bootstrap. Every false promotion, false block, abstention, candidate-caused failure, and clean false negative remains in its eligible denominator.

6 Results

A separately implemented checker recovered the same headline counts from the released safe aggregates. This provides local implementation separation, not external verification authority.

6.1 False promotion and clean coverage

The governed pipeline produced 0/360 false promotions. The strongest frozen comparator mechanism, ProvenAI-style citation-fidelity/influence, produced 180/360 (50.0%). The paired difference was -0.50 with 95% CI $[-0.553, -0.447]$, beyond the predeclared -0.05 practical margin. Figure 1 reports the full mechanism panel.

Both the governed pipeline and the strongest comparator promoted 60/60 clean positives, with zero clean false negatives. The clean-coverage difference and paired 95% interval were both exactly zero, satisfying the -0.05 non-inferiority margin. Figure 2 shows that every primary mechanism sits at 1.0 clean coverage; equal values are not visually jittered into a spurious frontier.

The registered undetermined-outcome contrast was not supported on this battery because the construction could not discriminate among systems. Every one of the eleven panel members, including the governed pipeline, correctly left all 30 insufficient-evidence cases undetermined, so the difference was zero with 95% CI $[0, 0]$.

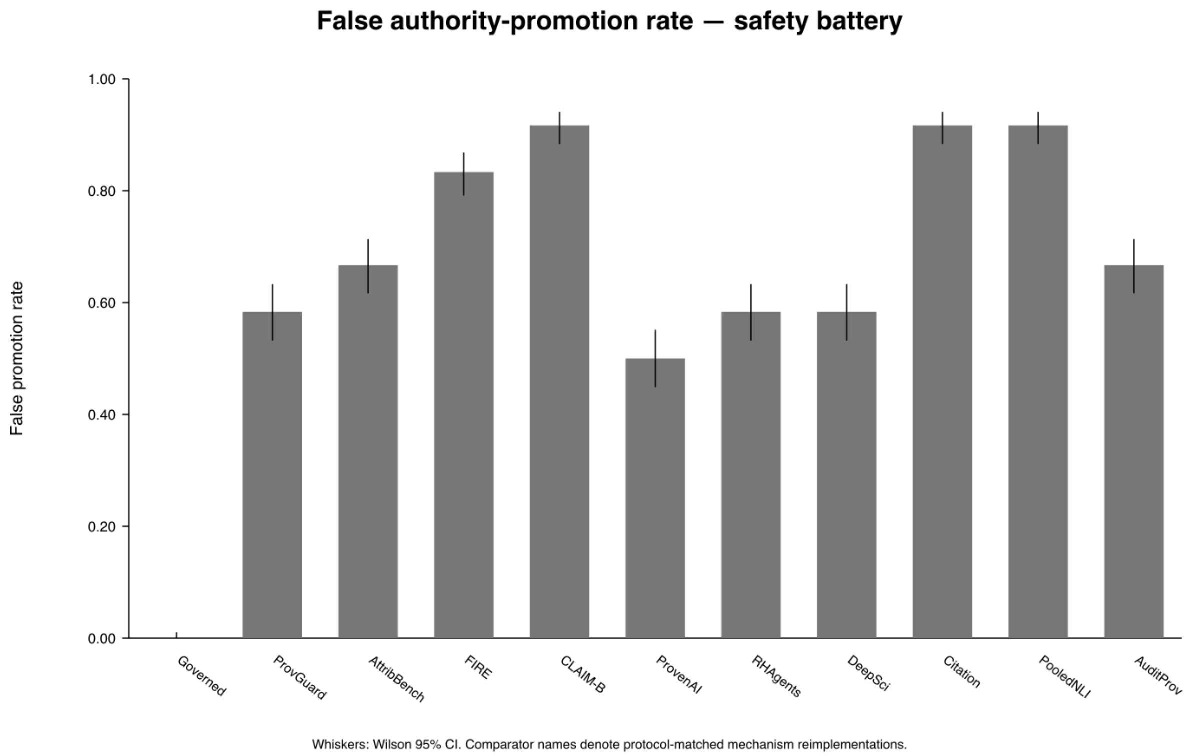

Figure 1: False authority-promotion rate over 360 hostile opportunities in the protected safety battery. Whiskers are Wilson 95% intervals for each rate. Comparator labels denote protocol-matched mechanism reimplementations, not the original external systems.

Table 2: Fail-closed outcomes and false-negative cost in the protected safety battery. The only gold family whose correct outcome is 'undetermined' contains 30 insufficient-evidence cases.

System	Correctly undetermined	False promotions	Clean FN
Governed pipeline	30/30	0/360	0/60
ProvenAI-style	30/30	180/360	0/60

The construction explains the saturation. The insufficient-evidence family had an *empty* evidence list, whereas every other family had exactly one evidence object. The rule "leave the claim undetermined if and only if the evidence list is empty" therefore classified all 420 cases correctly without scientific judgement. The comparison had no resolving power, so the null is retained as a negative measurement result rather than evidence that no difference exists or that abstention competence is equal. Table 2 reports the outcome and false-negative costs.

6.2 Shortcut-resistant exact-axis battery

A first repair removed the empty list but retained other shortcuts. Evidence-body length recovered the undetermined outcome on 20/20 protected cases with no false positives among the other 250, and the family name itself remained visible. Five of fourteen judgement-free probes recovered the undetermined label at informedness 1.00 on this construction; thirteen of fourteen did so on the original construction. Because repairing a named cue does not establish that no cue remains, no mechanism panel is reported for this intermediate design.

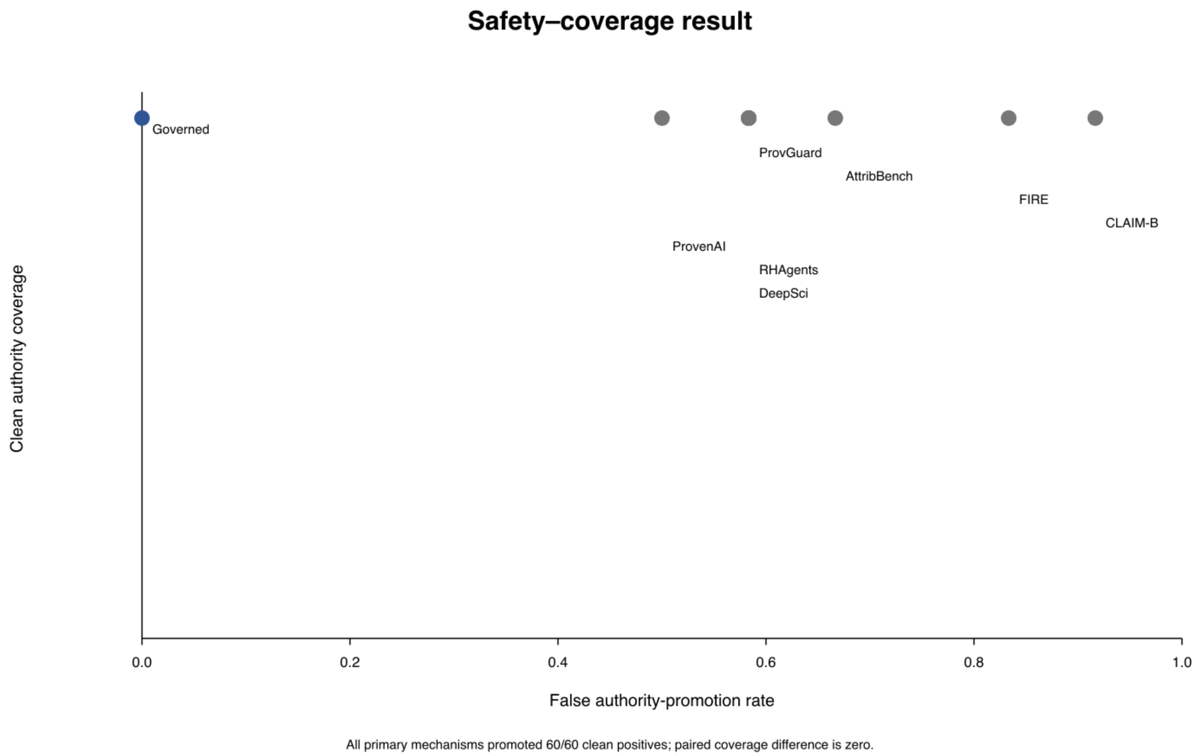

Figure 2: Safety–coverage result in the protected safety battery. Each point combines false-promotion rate over 360 hostile opportunities with promotion coverage over the same 60 clean controls. Every primary mechanism promoted 60/60 clean controls, giving zero paired coverage difference and a degenerate 95% interval at zero, while hostile false-promotion rates differ.

The final repair was specified by the target property before panel evaluation. All 840 evidence records use one shape-identical template with the same length, word count, punctuation, character-class profile, and container size. Support is carried by a token present on every record, so only token matching separates support from non-support. The undetermined family is split into two shape-identical subtypes and is a strict subset of a set that also contains 30 blocking cases. Fourteen judgement-free probes over counts, container lengths, key shapes, string lengths, character-class and template fingerprints, identifier shapes, and scalar values were fitted and evaluated before the mechanism panel. Across thirteen registered seeds, every probe had informedness 0.0 on the undetermined axis against the declared ceiling of 0.0; none predicted the undetermined outcome for any protected case.

On this shortcut-resistant battery, the governed pipeline selected the undetermined outcome on 30/30 eligible cases with 0/360 false promotions. The strongest frozen comparator selected by the primary false-promotion endpoint selected it on 0/30, giving a paired difference of 1.0 with 95% CI $[1.0, 1.0]$. The DeepSciVerify-style escalation mechanism selected it on 15/30, so the margin against that mechanism was 0.5, not 1.0. Nine of ten comparator mechanisms scored zero because they cannot express an undetermined outcome once the empty-evidence case is removed. The result therefore measures outcome expressiveness under the non-compensatory gate lattice, not finer-grained scientific judgement. Clean coverage remained 1.0 for all eleven systems, so that axis was still saturated. This distinct battery does not reinterpret the retained null from the primary safety battery.

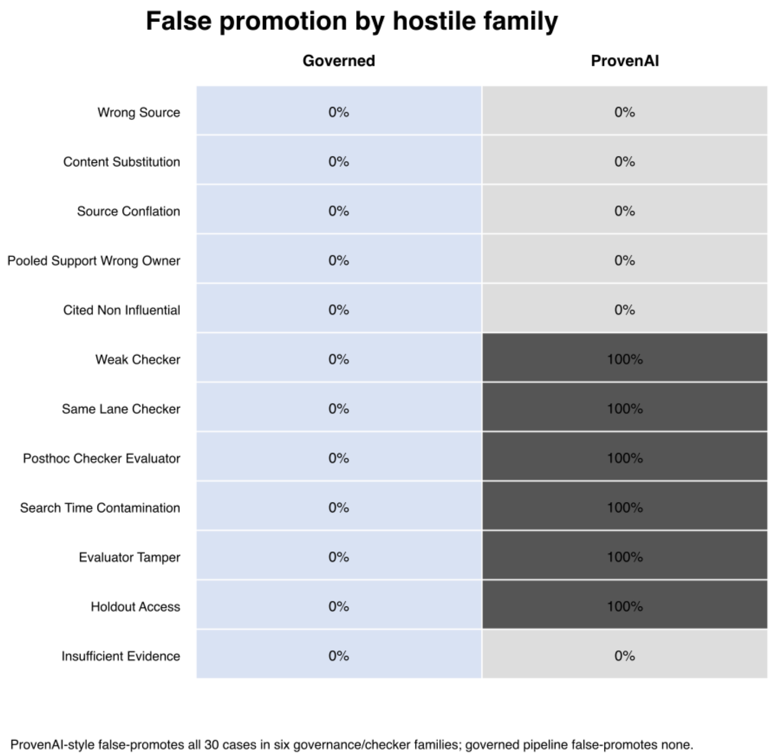

Figure 3: False-promotion rate by hostile family in the protected safety battery for the governed pipeline and the strongest frozen mechanism proxy. Each row contains 30 cases; the governed pipeline false-promoted none, whereas the proxy false-promoted all cases in six governance/checker families.

6.3 Attack families and ablations

The governed pipeline made no false promotion in any hostile family. The strongest comparator false-promoted all 30 cases in each of six governance/checker families: weak checker, same-lane checker, post-hoc checker/evaluator, search-time contamination, evaluator tamper, and holdout access. It was correct on content substitution, source conflation, wrong-source, pooled-owner, cited-non-influential, and insufficient-evidence families. Figure 3 shows the contrast.

Every registered ablation preserved 60/60 clean coverage while increasing false promotion. Removing exact content binding, checker-lineage independence, hostile-checker discrimination, behavioral influence, or the search-contamination block produced 30/360 false promotions (8.3%) each. Collapsing source/provenance identity produced 60/360 (16.7%); removing evaluator protection/telemetry produced 90/360 (25.0%); and replacing the fail-closed outcome with the predeclared six-of-nine soft-confidence threshold produced 330/360 (91.7%). Table 3 contains the complete panel.

6.4 Attribution, support, cost, and telemetry

Several mechanisms achieve perfect source-attribution accuracy while still false-promoting governance attacks; conversely, pooled semantic-support mechanisms can score well on support while attributing the source incorrectly. Figure 4 separates those coordinates.

Deterministic gate evaluation is slower than the simple mechanism proxies in this synthetic harness: mean latency is about 2.97 ms per case for the governed pipeline versus single-digit microseconds for the primary proxies (Figure 5). This is not an end-to-end serving benchmark; model and communication latency are intentionally absent.

Table 3: Protected safety-battery results. Comparator names denote protocol-matched mechanism reimplementations, not original external software. FP is over 360 hostile opportunities; clean is over 60 clean positives.

System / variant	FP	Clean	Clean FN	Latency (ms)	Resource
Governed pipeline	0.000	1.000	0	2.9736	9.0
ProvenanceGuard-style	0.583	1.000	0	0.0037	5.0
AttributionBench-style	0.667	1.000	0	0.0058	3.0
FIRE-style	0.833	1.000	0	0.0069	6.0
CLAIM-BENCH-style	0.917	1.000	0	0.0049	4.0
ProvenAI-style	0.500	1.000	0	0.0034	5.0
RewardHackingAgents-style	0.583	1.000	0	0.0072	5.0
DeepSciVerify-style	0.583	1.000	0	0.0074	6.5
Citation presence	0.917	1.000	0	0.0009	1.0
Pooled support	0.917	1.000	0	0.0049	2.0
Auditability/provenance	0.667	1.000	0	0.0033	5.0
Without exact content	0.083	1.000	0	2.7131	9.0
Without source/provenance	0.167	1.000	0	2.7077	9.0
Without checker lineage	0.083	1.000	0	2.7078	9.0
Without hostile checker	0.083	1.000	0	2.7043	9.0
Without influence	0.083	1.000	0	2.7045	9.0
Without evaluator protection	0.250	1.000	0	2.7116	9.0
Soft-confidence rule	0.917	1.000	0	2.4570	9.0
Without search block	0.083	1.000	0	2.7087	9.0

Process telemetry recorded zero protected-identifier hits and zero external network connections for both candidate and comparator mechanisms. Raw traces remain in protected custody. The separate checker recovered the same 0/360 versus 180/360 false-promotion counts and 60/60 versus 60/60 clean-promotion counts; this is local implementation separation rather than external result verification.

6.5 Exclusion of exploratory evidence

We exclude an earlier 39-case live-model arm from publication authorization. Post-evaluation audit found internally inconsistent live labels and expected outcomes, incorrect opportunity denominators in the summary calculation, and direct use of a candidate-hidden family label in the governed pipeline's decision logic. Those outputs remain adverse diagnostic evidence but are not combined with, substituted for, or used to tune the protected results.

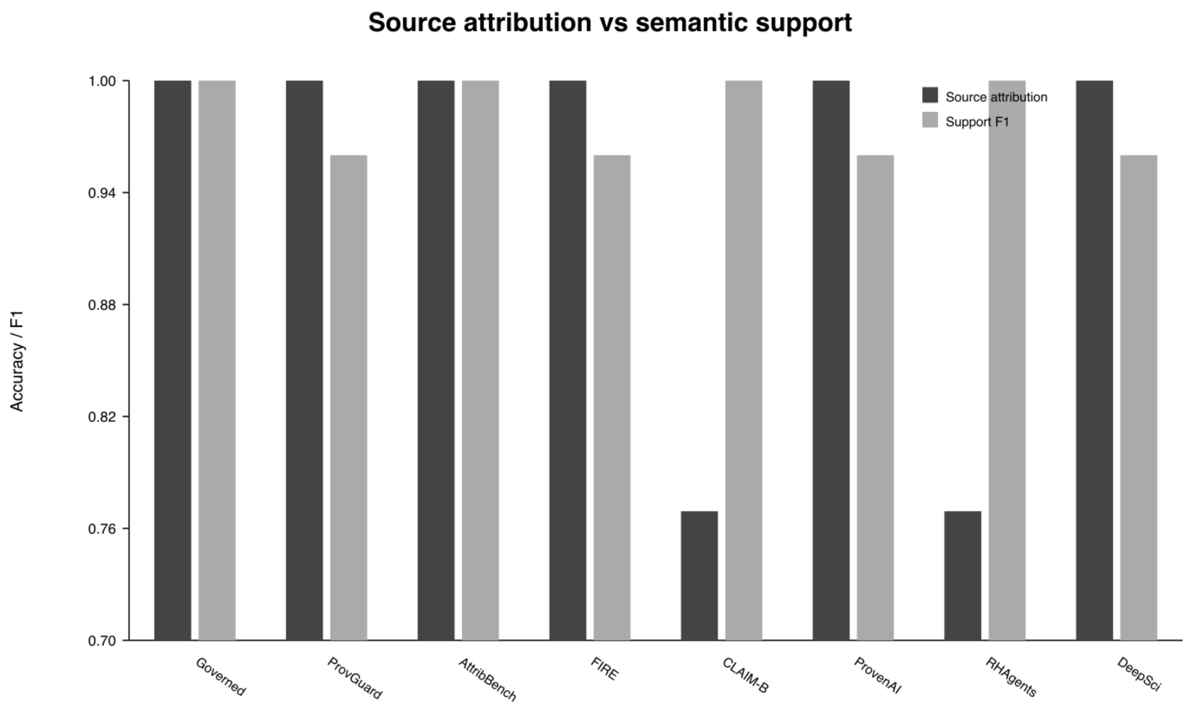

Figure 4: Source-attribution accuracy and support/contradiction F1 for the primary mechanism panel on the protected 420-case safety battery. Source attribution tests assignment to the correct evidence owner; support/contradiction F1 tests the semantic evidence label. High values on either coordinate do not guarantee a safe authority outcome.

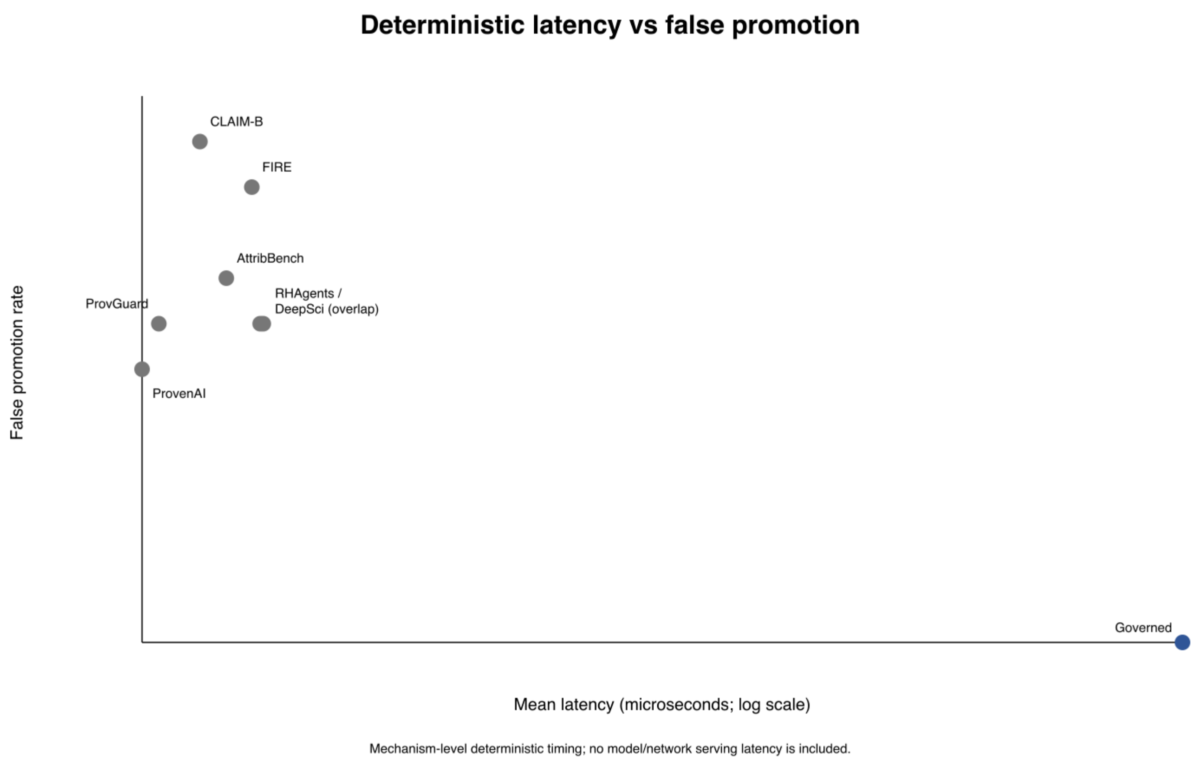

Figure 5: Mean deterministic evaluation latency across five repeats versus false authority-promotion rate over 360 hostile opportunities in the protected safety battery; the latency axis is logarithmic. Timing covers mechanism-level gate evaluation only and excludes model inference, communication, and end-to-end serving overhead.

7 Exact-contract extension: donor-complete scientific-promotion authority

The protected safety battery shows that a non-compensatory content, provenance, checker, and evaluator relation can eliminate false scientific-authority promotion relative to strong frozen mechanism proxies while preserving every clean positive. The exact-contract extension raises the comparison standard: ordinary provenance, claim verification, artifact/version binding, evaluator custody, epoch handling, and generic authorization are granted to a donor-complete product, leaving the final *scientific-promotion relation* as the discriminator. The safety battery and exact-contract extension remain separate prospectively frozen experiments and are not pooled.

7.1 Donor-engulfment architecture

The governed pipeline retains source-aware factuality, provenance, claim/evidence verification, citation fidelity, evaluator auditing, contamination defense, abstention, and generic authorization as donor-owned local components (Alvarez et al., 2026; Faizan & Alharthi, 2026; Liao, 2026; Tu et al., 2026; Zhou et al., 2026). The extension tests the higher-level authority relation that determines whether those successful local verdicts are sufficient to change the scientific standing of a target claim.

The target-bound relation is non-compensatory: a claim is promoted only when every registered hard scientific-promotion obligation is discharged. After ordinary provenance, verification, custody, and authorization checks have succeeded, the discriminating coordinates concern target-claim scope, independence of supporting evidence, and unresolved scientific-promotion authority.

7.2 Exact-contract design

The battery contains 400 protected cases spanning five heterogeneous scientific artifact domains, eight hostile/control archetypes, and ten protected variants per domain–archetype cell. All systems receive the same candidate-visible information. The four compared arms are:

- the target-bound non-compensatory scientific-promotion relation;
- a strong donor-complete generic-authorization product with ordinary provenance, verification, artifact/version, custody, and epoch information but without the complete target-bound relation;
- a compensatory all-signal product that permits successful coordinates to offset failed or unresolved ones; and
- an information-equivalent typed donor product given every target-bound scientific coordinate and the same promotion predicate, testing whether the semantics are portable.

7.3 Exact promotion decisions

The target-bound relation selects the correct scientific-promotion outcome on **400/400 protected cases**. The donor-complete generic product is correct on 250/400, the compensatory product on 50/400, and the information-equivalent typed product on 400/400 with no decision mismatch to the target-bound relation. The paired target-bound minus generic-product effect is +0.375, with a domain-stratified bootstrap 95% CI $[0.3275, 0.4225]$ and paired McNemar discordance of 150–0. The target-bound relation makes no false promotions, retains clean promotion at 1.0, and exceeds the generic product in every exact domain. A separately implemented checker recovered the same protected-row identity and all arm counts.

The result isolates one decision difference under the registered exact contracts. Provenance, verification, custody, and generic permission can all be valid while claim scope, evidence dependence, or an unresolved scientific obligation still prevents promotion. Making the target-bound relation explicit closes all 150 generic-product errors without sacrificing any clean promotion.

7.4 Portability result

The information-equivalent typed product also reaches 400/400 and is decision-identical to the target-bound relation. The promotion semantics are therefore sufficient and portable within the registered exact-contract representation: they can be implemented as a separate governed layer or as a correctly enriched donor product without changing any registered decision. The advantage over the donor-complete generic product identifies missing scientific coupling in that interface rather than an inherent centralization advantage.

7.5 Measurement-resolution audit

The extension does not reinterpret the earlier battery's secondary outcomes. Perfect clean coverage across all primary systems confirms that the false-promotion result is not blanket refusal, but it supplies no comparative ordering. The original insufficient-evidence construction is likewise saturated and remains a negative measurement result. These adverse outcomes stay visible while the exact-contract counts remain a distinct experiment.

7.6 Strongest supported statement

After donor-complete provenance, verification, custody, artifact binding, freshness, and generic authorization are granted, *scientific claim promotion remains a distinct non-compensatory authority relation in these registered exact contracts.* Making that relation explicit raises exact promotion accuracy from 0.625 to 1.000 against the strong donor-complete product while preserving zero false promotions and perfect clean coverage. Live verifier/provider transfer is the next empirical stress test of this bounded relation.

8 Threat model, limitations, and interpretation

The theory and the evidence have different scopes. Proposition 1 and its factorization corollary hold for finite exact-outcome decision problems, but they identify when a target is attainable from a representation and output alphabet; they do not show that a deployed verifier possesses a target-sufficient representation. The empirical layers establish only the frozen representations, relations, and case distributions evaluated here.

First, the primary safety and shortcut-resistant batteries each use 420 synthetic mechanical-gold cases rather than naturally occurring scientific disagreements. The written human-adjudication rubric was never triggered because all final cases were mechanically decidable, so no claim is made about expert agreement. The exact-contract extension broadens the evaluation to five artifact domains and eight hostile/control archetypes, but its 400 cases are still constructed contracts rather than a population sample of scientific disputes. The deterministic scoring logic isolates authority-transition semantics rather than the error distribution of a full stochastic research agent.

Second, the comparator mechanisms are matched reimplementations. They instantiate published ideas under one candidate-visible packet and common accounting, but the results are not a leaderboard against the original ProvenAI, ProvenanceGuard, DeepSciVerify, FIRE, AttributionBench, CLAIM-BENCH, Reward-HackingAgents, or related systems in their native environments. Likewise, 60/60 clean coverage shows that the governed pipeline did not obtain its false-promotion advantage by blanket refusal; these unambiguous controls do not estimate recall on difficult benign cases. The hostile taxonomy is finite, so adaptive attacks, compromised host infrastructure, unobserved side channels, and novel evaluator attacks remain outside the study.

Third, two registered axes in the primary safety battery are saturated and must be read as design limits rather than comparative findings. Clean coverage is 1.0 for all eleven primary systems and every registered ablation. The original undetermined-outcome family is also saturated because its empty evidence list makes every panel member correct on all 30 eligible cases. Thus the false-promotion endpoint supplies the comparative measurement, whereas the clean-coverage safeguard and undetermined-outcome null do not establish equality of underlying competences. The shortcut-resistant construction produces nonzero panel resolution, but nine of ten comparator mechanisms cannot emit the undetermined outcome. Its 1.0 margin against the strongest comparator and 0.5 margin against the escalation-capable mechanism therefore measure outcome/interface attainability under the frozen gate lattice, not general scientific judgement.

The nuisance audit has the same bounded scope. Fourteen frozen probes evaluated across thirteen registered seeds found no advantage on the shortcut-resistant undetermined axis. This is evidence against the declared shortcut class, not proof that no possible nuisance decoder exists. More generally, semantic attainability, nuisance resistance, matched outcome attainability, and panel resolution are separate requirements because failure of any one changes what a benchmark contrast can mean.

Fourth, the component mechanisms are not claimed as novel. Provenance tracking, attribution evaluation, citation checking, iterative verification, evidence escalation, contamination detection, evaluator-tamper detection, abstention, benchmark auditing, claim-level auditability, provenance ontologies, and generic behavioral-integrity checking all have prior art discussed in Section 2. The contribution tested here is their non-compensatory composition and the target-bound scientific-promotion relation above a donor-complete generic authorization product. The information-equivalent typed product's 400/400 tie is a positive boundary result: the evidence does not support a centralization advantage, universal necessity of the exact gate list, or superiority over deployed donor systems.

Finally, naturalistic transfer remains unestablished. A prospective programme specifies source–case-family clusters across four domains, paired identifiable controls, worst-domain analysis, source-disjoint replication, and independent evaluator custody. It has produced feasibility evidence but has not authorized a natural pair or executed the external comparator panel. These preflights define what a future naturalistic study would need but contribute no performance evidence to the present paper.

Computational integrity is also not scientific authority. Deterministic reconstruction, protected custody, and implementation separation can establish which computation was performed and whether it can be repeated under the released conditions. They do not establish that the benchmark measures the intended construct or that an external party has independently validated the scientific interpretation. A perfectly repeated flawed protocol would reproduce the flaw; this is why the saturated undetermined-outcome construction remains visible rather than being rewritten retrospectively.

9 Broader impact and governance

False scientific-authority promotion can amplify incorrect or misattributed claims, particularly when a system presents verification artifacts that appear independent but are not. A fail-closed layer may reduce this risk, but excessive blocking can delay legitimate scientific communication or concentrate power in whoever controls evidence and evaluator registries. Work on academic integrity and agentic abstention reinforces that refusal can be the correct behavior under genuine insufficiency, while also showing that productive abstention remains difficult and should not be conflated with generic non-response (Yang et al., 2026; Liu et al., 2026). We therefore report abstention cost, clean coverage, custody identity, and artifact transparency rather than optimizing only a security metric.

Protected evaluation creates another trade-off: some labels must remain hidden during scoring, yet publication requires enough evidence for audit. The review supplement therefore exposes safe aggregates, a post-evaluation public slice, and a separate-implementation check while protected holdout labels and raw traces remain in restricted custody. This supports local implementation separation, not external result-verifier custody.

10 Data and code availability

The anonymous review supplement contains safe aggregate result objects for the protected safety battery, the distinct shortcut-resistant exact-axis battery, and the 400-case exact-contract extension. It also contains checksums and a standalone verifier for the headline counts and retained interpretation boundaries. The supplement preserves the saturated undetermined-outcome comparison as not supported and keeps the three empirical layers separate rather than pooling them.

Protected per-case gold, raw execution traces, secret seeds, and candidate-hidden fields are withheld from the review supplement because they are part of the custody model under test. Their absence is explicit rather than treated as public reproduction. The supplied verifier therefore checks the published safe aggregates and headline counts; it neither reruns protected scoring nor reconstructs hidden gold from information unavailable to the candidate or comparator mechanisms.

For double-blind review, identifying ownership metadata and author-linked archive information are excluded from the manuscript-facing supplement. Its exact members are checksum-bound and can be uploaded as anonymous supplementary material. Persistent access and archival identifiers will be supplied in the non-blinded version without changing the scientific result; no persistent DOI is asserted at this review stage.

The naturalistic source-expansion programme is not part of the evidence supporting the present performance claims. Its outcome-free feasibility records remain separate so that an unexecuted successor is distinguishable from an omitted experiment.

11 Conclusion

A scientific-verification result requires more than a score. The intended outcome must be identifiable from admissible scientific information, resistant to the nuisance class being tested, expressible by the compared interfaces, and capable of varying over the panel. The exact-error and factorization results make these requirements explicit: a target relation is implementable when it factors through the decision representation and its required outcomes are available, whereas mixed fibres impose irreducible error.

The three empirical layers instantiate different parts of that theory without rewriting one another. The primary safety battery observed 0/360 false promotions for the governed pipeline versus 180/360 for the strongest frozen mechanism comparator while both promoted 60/60 clean positives; its original undetermined-outcome family was construction-saturated and remains not supported. The shortcut-resistant battery removed the registered cue class and separated the undetermined outcome, but because most comparators cannot emit that outcome, the supported result concerns interface/outcome attainability rather than general scientific judgement. The exact-contract extension then raised the comparison standard: after donor-complete provenance, verification, custody, version, epoch, and generic-authorization machinery were granted, the target-bound non-compensatory relation scored 400/400 exact decisions versus 250/400 for the donor-complete generic product and 50/400 for a compensatory product. An information-equivalent typed product also scored 400/400, showing that the semantics are portable rather than inherently centralized.

The contribution is therefore wider than a gate checklist but narrower than a deployed-system victory. It provides a theory for deciding when verification axes can support scientific claims and finite protected evidence showing that local verification and generic permission need not suffice for a target scientific-promotion relation. Naturalistic, source-disjoint, externally custodied evaluation remains open: the prospective source programme has not authorized a single natural pair or executed the external comparator panel. Those successor preflights are retained as prospective records, not promoted into evidence for the present paper.

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